

Research Statement

Stanford Summer Research Program -- Kavish Kumar

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I am a final-year undergraduate in Computer Science and Artificial Intelligence at Plaksha University, with a developing interest in applied machine learning. My coursework and project experience have given me a foundation in core ML methods, and the Stanford Summer Research Program represents an opportunity to work alongside researchers who can help me understand what rigorous research practice actually looks like in a leading environment.

My most substantial project involved building a speech recognition system for people with dysarthria -- a motor speech disorder associated with conditions like ALS and Cerebral Palsy. Standard ASR systems perform poorly on dysarthric speech because they are trained on neurotypical data, and I wanted to explore whether a fine-tuned wav2vec 2.0 model, paired with a bidirectional LSTM, could do better. Working with the TORGO dataset, I achieved a word error rate of 48% and a character error rate of 27%, which is reasonable for a speaker-independent setup -- though there is clearly significant room for improvement. The project taught me a great deal about the gap between getting a model to run and getting it to generalise.

I have also worked on more applied data science problems. I built a flight delay classification model using XGBoost and FlightRadar24 data, and in the summer of 2024 I participated in a short research project at NTU Singapore on box office prediction using k-means clustering and random forest regression. These projects were largely self-directed, which meant I learned to scope a problem practically, but also that I sometimes made choices I would now revisit with better guidance.

At Stanford, I am open to working on problems at the intersection of machine learning and either language or signal processing. I am most comfortable in Python and have worked with PyTorch and the HuggingFace library. I am not looking to arrive as an independent researcher -- I want to contribute to ongoing work, learn how a strong research group structures its process, and come away with a clearer sense of the problems I want to focus on going forward.

I am aware that my academic record is not exceptional, and I do not want to overstate what I bring. What I can offer is genuine interest, a willingness to work hard, and the technical grounding to be useful in a lab context from an early stage.